

The Paradox of an "Open" Internet and AI Realities in the Global Majority

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Introduction: the illusion of an open internet

The internet is often described as an open infrastructure: anyone with connectivity can theoretically access its networks and services. In practice, however, openness is deeply uneven. For billions of people across the Global Majority³, the experience of an open internet is fundamentally different and shaped by structural barriers that shape what information they can access, how they communicate, and whether they can meaningfully participate online. AI systems are now systematically encoding and amplifying these inequalities. Understanding how this transformation is occurring requires moving beyond the assumption that technical access alone guarantees openness. Instead, the emerging AI divide reveals how linguistic exclusion, unequal data representation, and the concentration of computational resources are reshaping who can participate in the digital ecosystem.

From the digital divide to the AI divide

The digital divide has traditionally been defined by disparities in internet access (Sambuli, 2016), high mobile data costs that are, on average, three times higher than in economically developed countries (GSMA, 2025), and limited internet infrastructure penetration (Okolo, 2023). While there has been significant progress over the past decade, African countries continue to struggle with reliable and affordable internet access (ITU & UNESCO, 2025). At the same time, governments, telecommunications companies, tech giants, and international development institutions have accelerated investments in digital infrastructures. Examples include the World Bank's financing of fibre-optic expansion in Nigeria (World Bank Group, 2025) and Meta's 2Africa subsea internet cable, constructed in collaboration with a consortium of leading telecom companies like Orange, MTN, and Vodafone (2Africa, 2022).

Increasingly, the digital divide is evolving into what policymakers describe as the AI divide (United Nations & ILO, 2024). Inequalities in internet access are now compounded by disparities in AI development, research capacity, and governance. This new divide and crisis in openness manifests in three interconnected dimensions. First, linguistic exclusion limits access to information in the languages spoken by much of the world's population. Second, epistemic erasure of culture and values due to data scarcity compounds challenges to linguistic inclusion. This leads to the marginalisation of

³ Global Majority includes communities in low and middle-income countries across Africa, Asia, Latin America, the Caribbean, and the Pacific Islands.

cultures and knowledge systems in AI models. Finally, the concentration of computational power in a small number of countries limits how countries, particularly those in the Global Majority, can contribute to the development of truly global AI systems.

Linguistic access and meaningful participation

Language remains one of the most significant barriers to meaningful participation online. When speakers of non-English and non-Romance languages attempt to access information or communicate online in their respective languages, they often encounter systematic barriers. These impact their ability to receive effective content moderation (Elsawah et al., 2025), achieve meaningful dissemination of their content (Graham, 2014), or maintain control over their intellectual property (Bhuiyan, 2014; Story et al., 2006).

AI systems amplify these inequities exponentially. Large language models perform significantly worse in low-resource languages (Li et al., 2025). Content moderation algorithms struggle to identify hate speech or misinformation in languages outside their training datasets while simultaneously over-flagging benign content because they lack cultural context (De Gregorio & Stremlau, 2023; Shahid & Vashistha, 2023). These issues cause recommendation systems to systematically deprioritise non-English content, creating algorithmic invisibility for billions of people who become more vulnerable online. In some cases, shortcomings in content moderation for low-resource languages amplify harmful content, further exacerbating internet challenges for millions of users (Nigatu & Raji, 2024).

More broadly, these challenges represent a fundamental barrier to Global Majority internet users' participation in the digital economy, access to knowledge, and the ability to shape public discourse. When openness is measured solely by the ability to connect to a platform rather than the ability to use it meaningfully, technical accessibility becomes a misleading indicator of meaningful participation.

Data colonialism and the homogenization of knowledge

A second challenge concerns whose knowledge, culture, and values are encoded in AI systems. Research from the Data Provenance Initiative suggests that training data from regions such as Africa and South America accounts for only approximately 0.2% of the datasets used to develop major machine-learning models (Longpre et al., 2024).

This lack of representative datasets constitutes a profound epistemic injustice. AI systems, particularly large language models and generative image systems, are gradually becoming authoritative sources of information (ARTICLE19, 2025) and redefining cultural narratives (Symons & Abumusab, 2024). Yet they homogenise thought, culture, and values into a mainstream ideal (Sourati et al., 2025). This homogenization predominantly reflects Western, and frequently American, perspectives, opinions, and aesthetics, further deprioritising the vast array of worldviews.

This dynamic mirrors historical patterns of colonial knowledge extraction and erasure, but with a crucial difference: it is being automated, scaled, and embedded into

infrastructure that increasingly mediates how people and communities access information and understand the world. The open internet promised democratisation of knowledge, but AI-driven concentration threatens to narrow whose knowledge counts as legitimate, whose culture deserves representation, and whose values shape the collective digital future. As AI tools become integrated into search engines, productivity platforms, and educational resources, they risk reinforcing a narrow set of cultural assumptions while sidelining diverse intellectual traditions.

The geography of computational power

A third and more structural challenge to openness lies in the geography of computational resources. The infrastructure required to train advanced AI systems—data centres, specialised graphics processors, and large-scale cloud platforms—is disproportionately concentrated in North America and Europe (Lehdonvirta et al., 2024). Research from the Oxford Internet Institute describes this AI compute inequality in three categories: the Compute North, Compute South and, Compute Desert. Wealthy countries in the Compute North host data centres equipped with the latest GPUs capable of training frontier AI models. These countries can participate in cutting-edge AI development, iterate on novel architectures, and maintain technological sovereignty. Countries in the "Compute South" are mainly middle-income countries with access to older GPU infrastructure, which is sufficient for deploying and fine-tuning existing models but insufficient for training frontier systems. These nations are relegated to being consumers and adapters of AI technology rather than originators. Meanwhile, countries in the Compute Desert have little or no access to AI computing infrastructure. These nations have no meaningful participation in the AI economy beyond basic usage of consumer applications.

These disparities create a structural hierarchy in the AI ecosystem. Notable compute shortages in Africa, Latin America, the Caribbean, and Oceania mean that these regions are systematically relegated to the bottom of the AI development hierarchy. This is not due to lack of talent, ambition, or potential applications, but rather structural inequalities in access to the fundamental infrastructure required for participation.

The compute divide between regions creates a vicious cycle, compounding challenges of both the AI and digital divides. Without access to computational resources, researchers and developers in these regions cannot train models on their own data or build systems tailored to local challenges, in their own languages. They become dependent on models developed elsewhere, trained on others' data, optimised for others' use cases (Chege, 2025). The promise of openness rings hollow when the basic infrastructure required for meaningful participation is so unevenly distributed.

Governing AI for genuine openness

The challenges posed by linguistic exclusion, data scarcity, and the consolidation of computational power require governance frameworks that move beyond expanding connectivity alone. Ensuring meaningful openness, transparency, and accountability in

the age of AI means addressing substantive participation throughout the entire value chain of AI development: who can build, train, and shape these systems.

First, research communities should advocate for linguistic equity, requiring AI systems operating globally to demonstrate meaningful performance across diverse language communities. This could trigger investments in low-resource language technologies, supporting community-driven dataset development, and establishing performance benchmarks that account for linguistic diversity.

Second, policymakers and non-governmental stakeholders should support frameworks for data sovereignty and epistemic justice that recognise the rights of different communities to shape how their knowledge, culture, and contexts are represented in training datasets. This might include requirements for geographic and cultural representation in training data, community-consent mechanisms for data collection, and support for locally developed models that reflect local contexts.

Third, governments, research institutions, and civil society organisations must invest in computational infrastructure as a global public good. These investments must also be accompanied by initiatives that democratise access to the computing power needed for AI development. This could involve distributed compute networks, subsidised access for researchers in under-resourced regions, and support for regional AI research centres.

Conclusion: reclaiming openness

The open internet was once envisioned as a democratizing force capable of expanding access to information and enabling new forms of global participation. In the age of artificial intelligence, however, openness cannot be defined by connectivity alone. True openness now depends on linguistic inclusion, equitable data representation, and access to the computational infrastructure required to build digital technologies.

Fortunately, many of the approaches described in the contribution are already underway. The newly launched Africa Green Compute Coalition (AGCC), led by UNDP, aims to democratise access to sustainable, affordable AI computing power across the African continent (UNDP, 2025). A growing number of national governments and regional initiatives across the Global Majority are developing open-source AI models. These include the development of LatamGPT led by Chile's National Center for Artificial Intelligence and a consortium of government, academic, and civil society partners (CENIA, 2026), the launch of the SEA-LION family of open-source LLMs that better understands Southeast Asia's diverse contexts (Ng et al., 2025), and Nigeria's N-ATLAS LLM, developed for 3 Nigerian languages and Nigerian pidgin English to increase access to government services (FMCIDE, 2025).

These efforts demonstrate that through coordinated action, meaningful inclusion, and local collaboration, the internet can begin to fulfil its long-deferred promise as a genuinely equitable resource shaped by and for the communities it has historically overlooked. By evolving into a more just and inclusive ecosystem, AI systems may one day reflect the full diversity of human experience rather than perpetuating the compounding barriers that have long defined access for the Global Majority.

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